WRITTEN ASSIGNMENT

Tasks for Course: DLMAIRIL01 – Reinforcement Learning

CONTENT

1. Task 2
1.1 Task 1: Alpha Go 2
1.2 Task 2: Playing Pong 2
1.3 Task 3: Neural Architectural Search 2
2. Additional information for the evaluation of the written assignment 3
3. Tutorial Support 3

1. TASK

Please choose one of the topics listed below to write your assignment on.

The starting point for your term paper will be the course book, the contents of which will serve as the basis for an in-depth examination of one of the following questions. You are expected to research and cite from sources corresponding to your chosen topic.

Note on copyright and plagiarism:

Please take note that IU Internationale Hochschule GmbH holds the copyright to the examination tasks. We expressly object to the publication of tasks on third-party platforms. In the event of a violation, IU Internationale Hochschule is entitled to injunctive relief. We would like to point out that every submitted written assignment is checked using a plagiarism software. We therefore suggest not to share solutions under any circumstances, as this may give rise to the suspicion of plagiarism.

1.1 Task 1: Alpha Go

AlphaGo reached and exceeded human-level performance in the game of Go, surprising even experts that this can be accomplished at the time. Describe how AlphaGo works, which deep neural networks are used for which task, and how deep learning approaches were combined with Monte Carlo Tree search to achieve the performance. Then compare the differences between AlphaGo and AlphaGo Zero.

1.2 Task 2: Playing Pong

Pong is a simple computer game imitating table tennis originally developed by Atari in 1972. It is now available as a simulation in OpenAI Gym. Devise a reinforcement learning approach using neural networks to learn how to play “pong” automatically. Include code examples and results where appropriate.

1.3 Task 3: Neural Architectural Search

Deep neural networks have shown remarkable performance in many areas such as image recognition and classification and many others. The performance of the deep neural networks critically depends on their architecture, i.e. which elements to use in which combination in which layer. Neural Architecture Search (NAS) aims at automating the search for the optimal network architecture. Explain how reinforcement learning can be used for this task.

2. ADDITIONAL INFORMATION FOR THE EVALUATION OF THE WRITTEN ASSIGNMENT

When conceptualizing and writing the written assignment, the evaluation criteria and explanations given in the writing guidelines should be considered.

3. TUTORIAL SUPPORT

In this Written Assignment task, several support channels are open; as the student, it is your responsibility to select your preferred support channel. The tutor is available for technical consultations and for formal and general questions regarding the procedure for processing the Written Assignment. However, the tutor is not required to approve outlines or parts of texts and drafts. Independent preparation is part of the examination work and is included in the overall evaluation. However, general editing tips and instructions are given in order to help you get started with the Written Assignment.